

Lab# 9: Use Cloud-based Malware Analysis Portals**#1 URL Analysis using ANY.RUN****Example #1: Command & Control Server URL**

Step 1: ThreatMiner → <https://threatview.io/> → Feeds → C2 Feed
<https://threatview.io/Downloads/High-Confidence-CobaltStrike-C2%20-Feeds.txt>

```
#Command and Control Feed Generated by Threatview[.]io Proactive Hunter on 03 July 2024. Domain and IP feed https[://]threatview[.]io/Downloads/High-C
# See any false positives ? Email us feeds[at]threatview[.]io
#IP,Date of Detection,Host,Protocol,Beacon Config,Comment
162.251.94.192,03 July 2024 10:16 AM UTC,qianxinnbplus.xyz,https,"www.qianxinnbplus.xyz/fam_cart.html",Generated by Threatview[.]io
147.45.178.94,30 June 2024 08:37 PM UTC,berjimek.com,https,"amateur-locket-gw.aws-usel.cloud-ara.tyk.io,/api/v2/login",Generated by Threatview[.]io
185.143.223.43,30 June 2024 08:37 PM UTC,londopas.com,https,"fortunate-homonym-gw.aws-eucl.cloud-ara.tyk.io,/api/v2/login",Generated by Threatview[.]io
91.238.181.235,16 June 2024 02:37 PM UTC,blmdiscount.com,https,"blmdiscount.com/Level/v3.82/1THMFWTJ38",Generated by Threatview[.]io
119.82.97.10,16 June 2024 02:37 PM UTC,iee-ece.info,https,"119.82.97.10/jquery-3.3.1.min.js",Generated by Threatview[.]io
89.23.108.208,16 June 2024 02:37 PM UTC,kauzalvip.com,https,"organic-satire-gw.aws-eucl.cloud-ara.tyk.io,/api/v2/login",Generated by Threatview[.]io
119.82.97.10,16 June 2024 02:37 PM UTC,nakit-yok.org,https,"119.82.97.10/jquery-3.3.1.min.js",Generated by Threatview[.]io
119.82.97.10,16 June 2024 02:37 PM UTC,nathanhr.services,https,"119.82.97.10/jquery-3.3.1.min.js",Generated by Threatview[.]io
```

Step 2: Login to ANY.RUN, paste a C2 link in ANY.RUN Sandbox Portal

New analysis Simple mode Pro mode New VM video streaming beta

URL or file upload

Type or copy URL
www.qianxinnbplus.xyz

Open in browser Download file and start

Start object from Temp directory Open in browser Internet Explorer

Operating system
Windows 10 (64 bit)

Auto confirm UAC On

Pre-installed soft set Complete Locale (OS Language) United States (en-US)

Step 3: Click 'Run Private analysis', notice the connections, process name, URLs spawned and content.

HTTP Requests	22	Connections	72	DNS Requests	108	Threats	0	Filter by PID, name or url	PCAP
Timeshift	Headers	Rep	PID	Process name	CN	URL	Content		
BEFORE	GET 200: OK	✓	2120	MoUsoCoreWorker.exe	Germany	http://www.microsoft.com/pkiops/crl/MicSe...	973 b ↓ binary		
3214 ms	GET 200: OK	✓	5344	firefox.exe	USA	http://detectportal.firefox.com/canonical.html	90 b ↓ text		
3936 ms	GET 200: OK	✓	5344	firefox.exe	USA	http://detectportal.firefox.com/success.txt?ip...	8 b ↓ text		
3989 ms	POST 200: OK	?	5344	firefox.exe	USA	http://o.pki.goog/s/wr3/XjA	83 b ↑ binary 471 b ↓ binary		

Step 4: Click IOC – Notice this domain is requesting DNS, IP & URL connections – Investigate!

IOCs

Summary of indicators of compromises 43

Copy selected

Main object – www.qianxinnbplus.xyz

? URL www.qianxinnbplus.xyz

Dropped executable file (1)

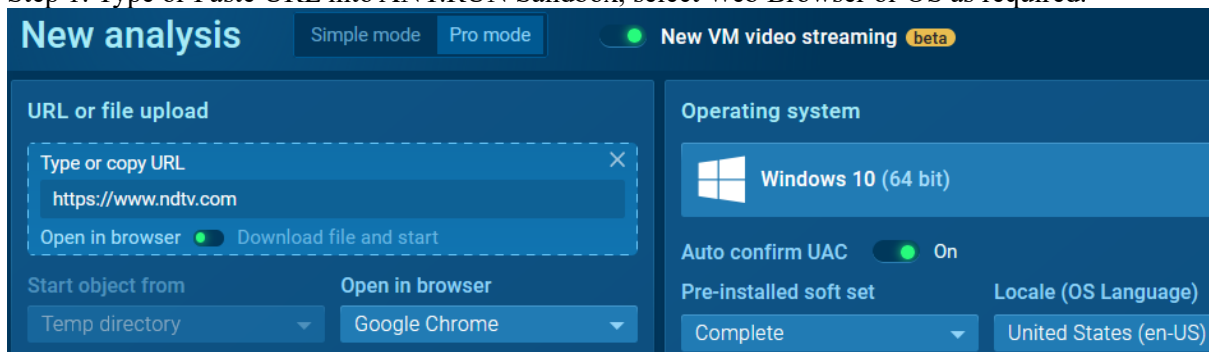
? SHA256 C:\Users\admin\AppData\Roaming\Mozilla\Firefox\Profiles\9kie7cg6.default-release\gmp-gmpopenh264\2.3.2\gmpopenh264.dll.tmp
d88dd3bfc4a558bb943f3caa2e376da3942e48a7948763bf9a38f707c2cd0c1c

DNS requests (10)		Connections (19)	
? DOMAIN	tiles-cdn.prod.ads.prod.webservices.mozgcp.net	? IP	34.160.144.191
? DOMAIN	r10.o.lencr.org	? IP	34.98.75.36
? DOMAIN	n.sni.global.fastly.net	? IP	2.16.202.112
? DOMAIN	www.lookfantastic.de	? IP	34.117.121.53
? DOMAIN	prod.ads.prod.webservices.mozgcp.net	? IP	95.101.54.211
? DOMAIN	prod.classify-client.prod.webservices.mozgcp.net	? IP	35.190.72.216
? DOMAIN	us-west1.prod.sumo.prod.webservices.mozgcp.net	? IP	34.149.100.209
? DOMAIN	r11.o.lencr.org	? IP	35.244.181.201
? DOMAIN	o.pki.goog	? IP	35.201.103.21
? DOMAIN	www.qianxinbplus.xyz	? IP	20.114.59.183

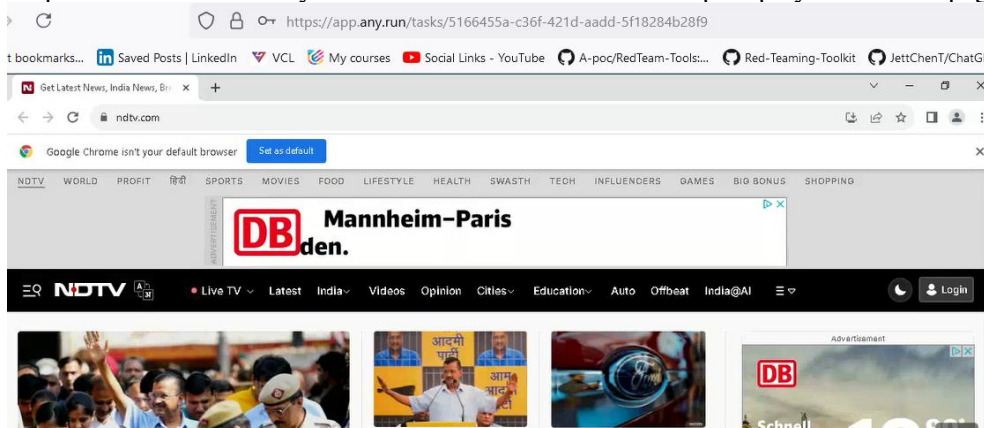
HTTP/HTTPS requests (12)	
? URL	http://www.microsoft.com/pkiops/crl/MicSecSerCA2011_2011-10-18.crl
? URL	http://detectportal.firefox.com/canonical.html
? URL	http://detectportal.firefox.com/success.txt?ipv4
? URL	http://o.pki.goog/s/wr3/XjA
? URL	http://r10.o.lencr.org/
? URL	http://o.pki.goog/wr2
? URL	http://r11.o.lencr.org/
? URL	http://ocsp.digicert.com/MFEWtZBNMEswSTAJBgUrDgMCGGUABBSAUQYBMq2awn1Rh6Doh%2FsBYgFV7gQUA95QNVbRTLtm8KPIGxvDl7190VUCEAJ0LqoXyo4hxxe7H%2Fz9DKA%3D

Example #2: News Portal (Good Web site) → <https://www.ndtv.com>

Step 1: Type or Paste URL into ANY.RUN Sandbox, select Web Browser or OS as required.



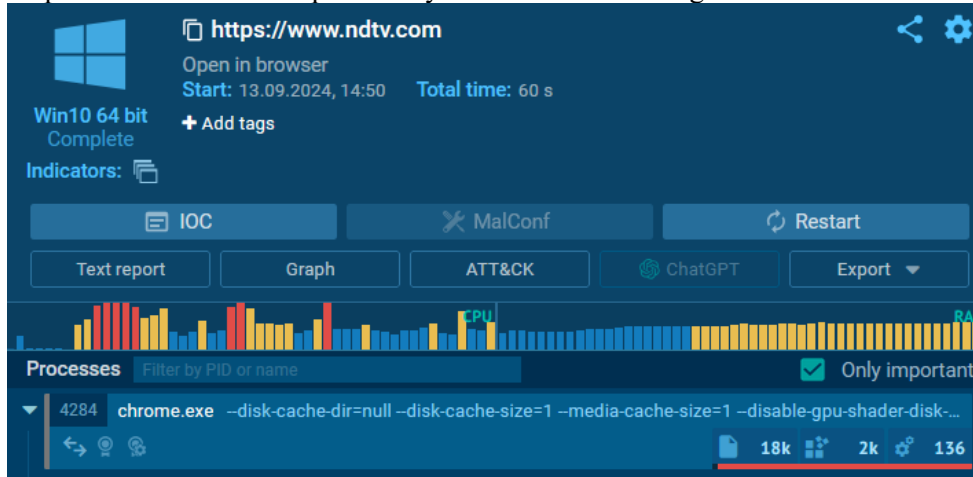
Step 2: Run Private Analysis and check the sandbox desktop display the default page for NDTV.



Step 3: Check HTTP Requests, Connections, DNS requests and Threats!

HTTP Requests	4	Connections	267	DNS Requests	433	Threats	2	Filter by PID, name or url
Timeshift	Headers	Rep	PID	Process name	CN	URL	Content	
BEFORE	GET 200: OK	?	-	-	🇩🇪	http://www.microsoft.com/pkiops/crl/MicSe...	973 b ↓ binary	
9088 ms	GET 200: OK	✓	6908	svchost.exe	🇺🇸	http://ocsp.digicert.com/MFEwTzBNMEswS...	471 b ↓ binary	
30564 ms	GET 200: OK	✓	7012	SIHClient.exe	🇩🇪	http://www.microsoft.com/pkiops/crl/Micros...	419 b ↓ binary	
30566 ms	GET 200: OK	✓	7012	SIHClient.exe	🇩🇪	http://www.microsoft.com/pkiops/crl/Micros...	407 b ↓ binary	

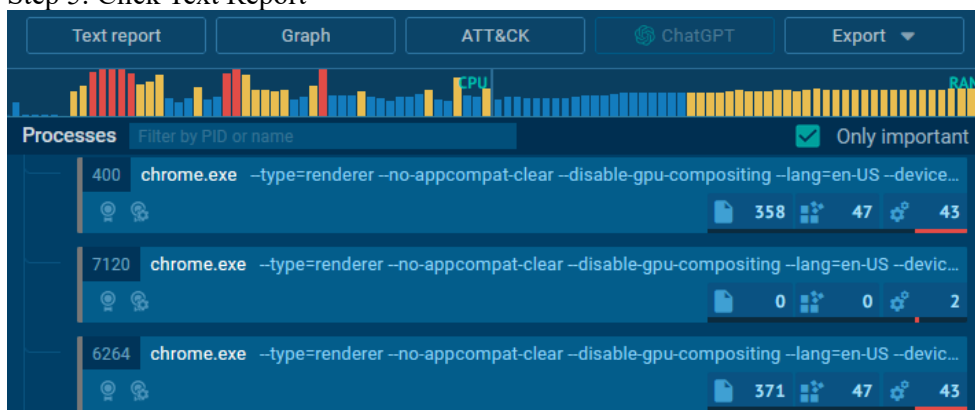
Step 4: Check Processes spawned by the browser accessing the site



IOCs	Summary of indicators of compromises	241
☒	☐ Copy selected	
Main object – https://www.ndtv.com		
? URL	https://www.ndtv.com	
DNS requests (69)		
? DOMAIN	t.hspvst.com	
? DOMAIN	ghb.adtelligent.com	
? DOMAIN	pm-widget.taboola.com	
👁️ DOMAIN	sb.scorecardresearch.com	
? DOMAIN	am-trc-events.taboola.com	
? DOMAIN	config.aps.amazon-adsystem.com	
? DOMAIN	invstatic101.creativecdn.com	
? DOMAIN	rtbpass-us.andbeyond.media	

Connections (167)
? IP 185.239.172.170
? IP 162.55.233.29
? IP 185.83.71.234
? IP 141.95.33.120
? IP 77.245.57.72
? IP 169.150.255.180
? IP 51.89.9.254
? IP 104.22.31.209
? IP 23.32.100.25
? IP 104.21.48.215
? IP 79.127.227.46
? IP 162.19.138.117
? IP 130.211.44.5

Step 5: Click Text Report



Example #3: Blocked URL → <http://117.202.112.174:33556/bin.sh>

Step 1: Threat Miner → <https://threatview.io/Downloads/URL-High-Confidence-Feed.txt>

Step 2: Paste the link in ANY.RUN

New analysis Simple mode Pro mode New VM video streaming **beta**

URL or file upload

Type or copy URL
`http://117.202.112.174:33556/bin.sh`

Open in browser ☒ Download file and start ☐

Start object from Temp directory Open in browser Internet Explorer

Change extension to valid ☒ On ☐ Hide source

Operating system

Windows 10 (64 bit)

Auto confirm UAC ☒ On

Pre-installed soft set Complete Locale (OS Language) United States (en-US)

Step 3: Click 'Run Private Analysis' and investigate

HTTP Requests		4	Connections		52	DNS Requests		40	Threats		0	Filter by PID, name or url	
Timeshift	Headers		Rep	PID	Process name		CN	URL		Content			
8388 ms	GET 200: OK		✓	6612	svchost.exe		🇩🇪	http://www.microsoft.com/pkiops/crl/MicSec...		973 b ↓ binary			
19510 ms	GET 200: OK		?	1332	svchost.exe		🇺🇸	http://ocsp.digicert.com/MFEwTzBNMEswS...		471 b ↓ binary			
28717 ms	GET 200: OK		✓	3568	SIHClient.exe		🇩🇪	http://www.microsoft.com/pkiops/crl/Micros...		419 b ↓ binary			
28718 ms	GET 200: OK		✓	3568	SIHClient.exe		🇩🇪	http://www.microsoft.com/pkiops/crl/Micros...		407 b ↓ binary			

Win10 64 bit Complete

`http://117.202.112.174:33556/bin.sh`

Open in browser
Start: 13.09.2024, 15:20

00:06

CPU 7% RAM 43%

Processes Filter by PID or name ☒ Only important

072	iexplore.exe	"http://117.202.112.174:33556/bin.sh"	1k	654	75
3660	msedge.exe	--from-ie-to-edge=8 - "http://117.202.112.174:33556/bin.sh"	17k	1k	155
6188	msedge.exe	--type=crashpad-handler "--user-data-dir=C:\Users\admin\AppData\Local\Micro...	65	13	25

HTTP/HTTPS requests (4)	
? URL	http://www.microsoft.com/pkiops/crl/MicSecSerCA2011_2011-10-18.crl
? URL	http://ocsp.digicert.com/MFEwTzBNMEswSTAIBgUrDgMCGgUABBSAUQYBMq2awn1Rh6Doh%2FsBYgFV7gQUA95QNVbRTlm8KPiGxvDL7I90VUCEAJ0LqoXyo4hxxe7H%2Fz9DKA%3D
? URL	http://www.microsoft.com/pkiops/crl/Microsoft%20ECC%20Product%20Root%20Certificate%20Authority%202018.crl
? URL	http://www.microsoft.com/pkiops/crl/Microsoft%20ECC%20Update%20Secure%20Server%20CA%202.1.crl

IOCs	
Summary of indicators of compromises 20	
Copy selected	
Main object – http://117.202.112.174:33556/bin.sh	
? URL	http://117.202.112.174:33556/bin.sh
DNS requests (2)	
? DOMAIN	edge-mobile-static.azureedge.net
? DOMAIN	edge-consumer-static.azureedge.net
Connections (13)	
? IP	117.202.112.174
? IP	40.126.32.68
? IP	13.107.246.42
? IP	13.107.21.239
? IP	40.68.123.157
? IP	38.343.36.151

- **File Analysis using ANYRUN**

Step 1: Use your VM (Remnux or Kali Linux)

Step 2: Download ZIP <https://github.com/Endermanch/MalwareDatabase/tree/master/ransomwares>

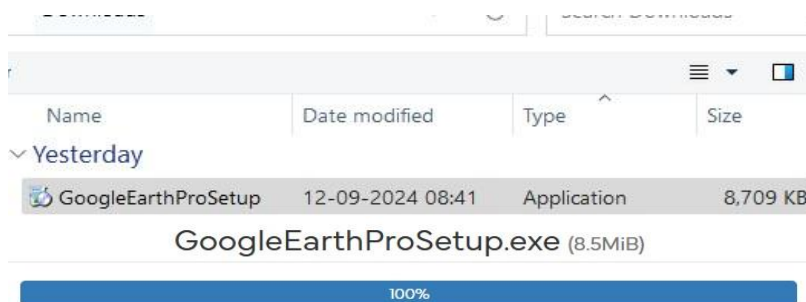
Step 3: Upload to ANYRUN, analyse.

#2 Hybrid-Analysis

Link → <https://www.hybrid-analysis.com/>



Step 1: Upload a file



Step 2: Select environment

Analysis Environments

Name

GoogleEarthProSetup.exe

Size

8.5MiB

Type

peexe

executable

MIME

application/x-dosexec

SHA256

e51043617f87a9...dde79f81e2d30

Available:

☒

Windows 10 64 bit

☐

Windows 11 64 bit

☐

Windows 7 32 bit

☐

Windows 7 32 bit (HWP Support)

☐

Windows 7 64 bit

☐

Linux (Ubuntu 20.04, 64 bit)

«

Back

Runtime Options

Generate Public Report

Step 3: Generate Public Report

Analysis Overview

Request Report Deletion

Submission

name:

GoogleEarthProSetup.exe

Size:

8.5MiB

Type:

peexe

executable

Mime:

application/x-dosexec

SHA256:

e51043617f87a960fbf8687115fc553f3b75d79fe61359e58f9dde79f81e2d30

Operating System:

Windows

Last Anti-Virus Scan:

09/13/2024 00:28:48 (UTC)

Last Sandbox Report:

09/13/2024 00:28:41 (UTC)

suspicious

Threat Score: 39/100

AV Detection: Marked as clean

#evasive

X Post

Link

E-Mail

Step 4: View report

<https://www.hybrid-analysis.com/sample/e51043617f87a960fbf8687115fc553f3b75d79fe61359e58f9dde79f81e2d30>

Step 5: Check Assessment report

Risk Assessment

Spyware

Found a string that may be used as part of an injection method

Persistence

Writes data to a remote process

Fingerprint

Queries process information

Evasive

Found a reference to a WMI query string known to be used for VM detection

Marks file for deletion

The input sample contains a known anti-VM trick

Writes archive files

Network Behavior

Contacts 1 domain and 2 hosts. View all details

DNS Requests

Login to Download DNS Requests (CSV)

Domain	Address	Registrar	Country
clients2.google.com	142.250.189.206 TTL: 300	MarkMonitor, Inc. Organization: Google Inc. Name Server: NS1.GOOGLE.COM Creation Date: 1997-09-15T00:00:00	United States

#3 FileScan.IO

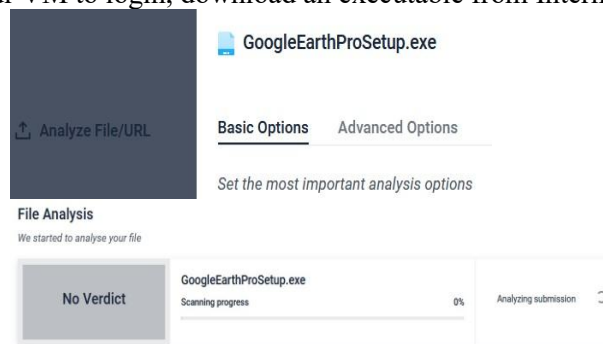
Link → <https://www.filescan.io/scan>

You can use the cloud-based portal to analyse files or perform Threat Hunting.

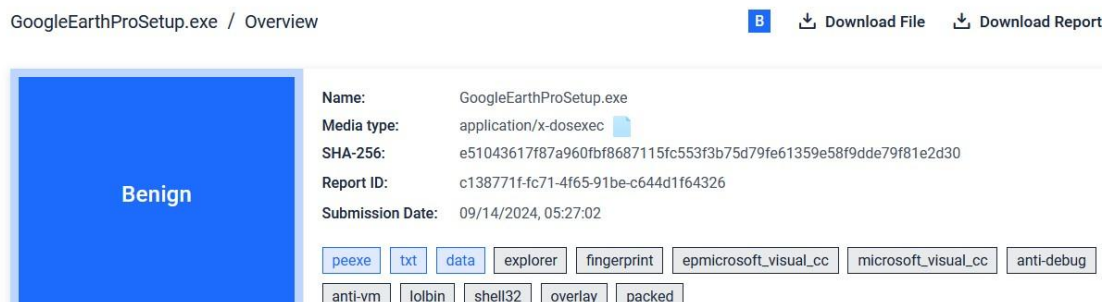
Step 1: Create your login account



Step 2: Use your VM to login, download an executable from Internet and upload a sample file



Step 3: Analysis overview



Step 4: File details

File Details

FileMagicDescription:	PE32 executable (GUI) Intel 80386, for MS Windows
Size:	8.50 MB
Architecture:	32 Bits binary
SubsystemReadable:	IMAGE_SUBSYSTEM_WINDOWS_GUI
Date:	Mon Aug 26 03:02:15 2024
Packers (PEID):	Safeguard v1.03 -> Simonzh
Compilers (PEID):	VC8 -> Microsoft Corporation
Linkers (DiE):	Microsoft Linker(14.0)[GUI32,signed]
Compilers (DiE):	EP:Microsoft Visual C/C++(2017 v.15.5-6)[EXE32], Microsoft Visua... (84)
Sign tools (DiE):	Windows Authenticode(2.0)[PKCS #7]
IsDigitallySigned:	true
IsDotNet:	false
IsPacked:	true

Step 5: ChatGPT Summary

ChatGPT Summary

GENERATED BY CHATGPT
14 Sep 2024 05:55

Executive Summary:

The analyzed malware has the following top five technical capabilities:

- **Command and Scripting Interpreter:** The malware is capable of taking command-line arguments, indicating its capability aligns with MITRE ATT&CK Technique T1059.
- **Thread Manipulation:** The malware imports APIs related to creating and terminating threads, suggesting system-level interaction.
- **Resource Interaction:** The malware imports APIs used to interact with resources, which may indicate its ability to access files, registry, and network.
- **Memory Manipulation:** The malware imports APIs used for memory manipulation, indicating its ability to modify memory. This capability can be leveraged for various malicious activities.

Step 6: IoCs

Indicators of Compromise (IOCs)

IOCs are extracted from the input binary or derived data (e.g. extracted files) of the analysis. Indicators that are highly likely to be an IOC are marked as interesting.

Generate IOC Query

Prevalence check 7 days

GoogleEarthProSetup.exe

Extracted Files

☒ Urls	116	
☒ Domains	114	
☒ IPs	37	

Any.Run Sample Reports:

- PDF → <https://any.run/report/283dcfaf32f981fc8a2285fc1399274cbd66ffa58683993ea979efe32534e5d3/5f6d4618-0eed-4805-adb4-4c2d73eba15c>
- MSI → <https://any.run/report/7d5e85dbdbf85ed033be48f7ef38ef438be15db869b2950a359f9e23cc1f58cb/9d14f3a5-cb5f-4836-b195-9d34a2347852>
- RAR → <https://any.run/report/57e9e0aba106f540c91c7864c22855a3f3a3e0a2b492926462d8ee1a8b6a754e/109ae2c2-c74d-4328-a97f-5a3284192485>

Lab activities:

- Use your VM to download at least three malicious IPs, Domains, URLs, ZIP/files.
- Study the ANY.RUN report for malicious files and present your summary.
- Find two Domains, URLs and IP Address - Good and malicious (from Threat Miner).
- Use your VM to download three different malicious ZIP and .EXE files each.
- Upload these artifacts to ANY.RUN (IP, Domains, URL, Files) and analyse to prepare your malware report based on the findings.